KAIZEN MATHS

Lesson starter

Bracket Expansion

Special Products and Advanced

Name _____

S1

$(x-5)^2$

S2

$(x+4)(x-4)$

S3

$2(x+3)(x-3)-5$

S4

$(x+1)(x+3)(x-1)$

Show your working

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ALG-3E2E-L3 · S

Show your working

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Exit ticket

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Name _____

E1

$(x+2)(x-2)(x-3)$

E2

$(x-2)^2$

E3

$(x+7)(x-7)$

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$(x+2)(x-2)(x-3)$

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$(x-2)^2$

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Teacher answers & working

Bracket Expansion · Special Products and Advanced

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S1

$(x-5)^2$

Answer:

$x^2-10x+25$

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Use the perfect square pattern.

$(x-5)^2=x^2-10x+25$

Step 3: Collect any like terms and write the expression in expanded form.

S2

$(x+4)(x-4)$

Answer:

x^2-16

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Recognize the brackets as $(A+B)(A-B)$.

$A=x,\quad B=4$

$A+B=x+4,\quad A-B=x-4$

$(A+B)(A-B)=A^2-B^2$

$(x+4)(x-4)=x^2-4^2$

x^2-16

Step 3: Collect any like terms and write the expression in expanded form.

S3

$2(x+3)(x-3)-5$

Answer:

$2x^2-23$

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Expand the product then multiply by the scale factor.

$(x+3)(x-3)=x^2-9$

$2x^2-23$

Step 3: Collect any like terms and write the expression in expanded form.

Teacher answers & working

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S4

$(x+1)(x+3)(x-1)$

Answer:

x^3+3x^2-x-3

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Expand two brackets then multiply by the third.

$(x+1)(x+3)=x^2+4x+3$

x^3+3x^2-x-3

Step 3: Collect any like terms and write the expression in expanded form.

E1

$(x+2)(x-2)(x-3)$

Answer:

$x^3-3x^2-4x+12$

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Expand two brackets then multiply by the third.

$(x+2)(x-2)=x^2-4$

$x^3-3x^2-4x+12$

Step 3: Collect any like terms and write the expression in expanded form.

E2

$(x-2)^2$

Answer:

x^2-4x+4

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Use the perfect square pattern.

$(x-2)^2=x^2-4x+4$

Step 3: Collect any like terms and write the expression in expanded form.

Teacher answers & working

Bracket Expansion · Special Products and Advanced

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E3

$(x+7)(x-7)$

Answer:

x^2-49

Working:

Step 1: Identify the expansion method and keep the brackets intact until each multiplication has been shown.

Step 2: Recognize the brackets as $(A+B)(A-B)$.

$A=x, \quad B=7$

$A+B=x+7, \quad A-B=x-7$

$(A+B)(A-B)=A^2-B^2$

$(x+7)(x-7)=x^2-7^2$

x^2-49

Step 3: Collect any like terms and write the expression in expanded form.